AI-Powered Multi-Agent Cybersecurity Framework for Smart City Lighting Systems

Ronak Ajwani, Shreya Chhatwani, Abdul Hannan Chougle, Himanshu Menghrajani, Dr. Mrs. Nupur Giri
Department of Computer Engineering, Vivekanand Education Society Institute of Technology, Chembur, Mumbai, India

***Abstract*—Smart city infrastructures, particularly IoT-enabled lighting systems, are increasingly vulnerable to sophisticated cyber threats such as Distributed Denial-of-Service (DDoS) attacks and malware intrusions. Traditional security approaches, which rely on static and signature-based detection mechanisms, struggle to adapt to dynamic, multi-vector, and large-scale attacks. To address these limitations, this paper proposes an AI-powered multi-agent cybersecurity framework for smart city lighting systems, leveraging real-time data streaming and large language model (LLM)-based threat intelligence. The proposed system adopts a zone-based architecture, categorizing city regions into critical, residential, and commercial zones, and deploys specialized agents for autonomous threat detection. Two primary agents—a DDoS Detection Agent and a Malware Detection Agent—operate in parallel using a five-layer pipeline comprising data sources, streaming (Apache Kafka), preprocessing, agent-based detection (LangGraph orchestration), and response layers. The agents utilize Groq-powered LLMs for contextual threat classification and automated mitigation. Experimental evaluation across multiple LLMs, including LLaMA, Mistral, and Gemma variants, demonstrates high detection performance, achieving up to 93.8% accuracy and 96.6% F1-score with low detection latency (28 seconds). The results highlight the effectiveness of combining multi-agent architectures with LLM-driven analysis for scalable, real-time cybersecurity in smart city environments.**

I. INTRODUCTION

The rapid advancement of smart city technologies has led to the widespread deployment of interconnected Internet of Things (IoT) systems, enabling efficient management of urban infrastructure such as transportation, energy, and public lighting. Among these, smart lighting systems play a crucial role in improving energy efficiency, reducing operational costs, and enhancing public safety. However, the increasing reliance on interconnected devices and real-time communication networks has significantly expanded the attack surface, making smart city infrastructures highly vulnerable to sophisticated cyber threats.

Cyberattacks such as Distributed Denial-of-Service (DDoS) attacks and malware intrusions pose serious risks to the availability, integrity, and reliability of smart city systems. Traditional cybersecurity mechanisms, which are primarily signature-based and static in nature, are often inadequate in detecting emerging threats such as zero-day attacks and multi-vector intrusions. Moreover, the large-scale and heterogeneous nature of IoT environments makes centralized monitoring and response mechanisms inefficient and difficult to scale. These challenges necessitate the development of intelligent, adaptive, and autonomous security solutions capable of operating in real-time.

Recent advancements in artificial intelligence, particularly the use of Large Language Models (LLMs), have opened new possibilities for intelligent threat detection and classification. However, applying such models directly to cybersecurity tasks requires domain-specific adaptation and integration with real-time data pipelines. In addition, the complexity of smart city environments calls for distributed and collaborative approaches rather than monolithic systems.

To address these challenges, this paper proposes an AI-powered multi-agent cybersecurity framework designed specifically for smart city lighting systems. The proposed approach models the city as a set of distinct zones, including critical, residential, and commercial areas, enabling context-aware monitoring and threat analysis. A multi-agent architecture is employed, consisting of specialized agents such as a DDoS Detection Agent and a Malware Detection Agent, which operate in parallel to analyze streaming data from IoT devices and network events. The system leverages real-time data streaming using Apache Kafka, agent orchestration through LangGraph, and LLM-based threat classification using Groq-powered models to enable efficient and scalable threat detection.

The key contributions of this work include the design of a zone-based smart city security model, the development of a multi-agent framework for parallel threat detection, and the integration of LLM-driven intelligence for enhanced classification and response. Experimental results demonstrate that the proposed system achieves high detection accuracy with low latency, making it suitable for real-time deployment in smart city environments.

II. RELATED WORK

Recent advances in artificial intelligence and large language models have significantly shaped the cybersecurity landscape, particularly in multi-agent systems, threat detection, and smart city security. The following studies reflect the current state of research in these interconnected domains.

Reference [1] explores the implementation of multi-agent AI systems as a coordinated cybersecurity solution for smart cities, leveraging autonomous agents capable of perceiving, learning, and responding to threats across critical urban sectors such as transportation, healthcare, and energy infrastructure. The study demonstrates improved scalability and response quality as agent counts scale from 50 to 200, validating the effectiveness of decentralized coordination protocols combined with LLM-driven decision-support modules. However, the

work remains largely simulation-based and does not address real-world deployment constraints or adversarial robustness.

Reference [2] investigates the application of LLM agents for automating threat detection and response, demonstrating their ability to review large volumes of security data, identify behavioral patterns, and generate contextually appropriate responses. The paper highlights that LLM agents bridge the gap between emerging threats and static security systems by employing contextual anomaly detection superior to traditional statistical methods. Key limitations include susceptibility to adversarial attacks on the LLM itself, high computational costs, and unresolved challenges around interpretability and prompt injection risks.

Reference [3] proposes CyKG-RAG, a novel framework that integrates cybersecurity knowledge graphs with Retrieval-Augmented Generation (RAG) to improve threat detection and analysis. By leveraging rich semantic relationships and structured data within knowledge graphs, the system provides contextually relevant information to LLMs, addressing hallucination and knowledge deficiency issues inherent in naive RAG approaches. Validation through real-world use cases demonstrates promising results; however, the dependency on pre-constructed knowledge graphs limits adaptability to rapidly emerging threat types not yet captured in the graph.

Reference [4] presents a comprehensive study on cybersecurity threats facing smart cities, surveying the security challenges across key urban domains including smart transportation, healthcare, environment, and public services. The paper provides a detailed taxonomy of attack types and defense countermeasures, and introduces an intrusion detection-based approach tailored for smart city infrastructure. While thorough in scope, the study acknowledges that existing intrusion detection mechanisms often lack the contextual intelligence needed to respond to the sophisticated and evolving nature of modern smart city threats.

Reference [5] introduces a multi-agent system for cybersecurity threat detection and correlation using LLMs, presenting a modular architecture where a task dispatcher agent coordinates domain-specific agents handling email analysis, log inspection, and network traffic monitoring. The system achieves high classification accuracy by restricting LLMs to structured semantic modules rather than treating them as fully autonomous agents, improving interpretability and reproducibility. Limitations include reliance on predefined analytical workflows that may reduce flexibility when confronting entirely novel attack patterns.

Reference [6] presents MALCDF, a distributed multi-agent LLM framework for real-time cyber defense, comprising four specialized agents — Detection, Intelligence, Response, and Analysis — that communicate over a Secure Communication Layer with encrypted, ontology-aligned messaging. Evaluated on a CICIDS2017-derived stream, MALCDF achieves 90.0% detection accuracy and an 85.7% F1-score with 6.8 seconds average per-event latency, outperforming both a lightweight random forest baseline and a single-LLM setup. However, the evaluation is limited to a small 50-record stream, raising questions about scalability and generalizability to large-scale production environments.

Reference [7] introduces Chimera, the first LLM-based multi-agent framework for automatically simulating both benign and malicious insider activities in enterprise environments, generating comprehensive system logs across six modalities including email, web browsing, file operations, network traffic, and system calls. Accepted at NDSS Symposium 2026, the framework produces ChimeraLog, a realistic benchmark dataset that significantly challenges existing insider threat detection models compared to prior synthetic datasets such as CERT. Despite its contributions, Chimera's simulation fidelity is constrained by the availability of accurate organizational profiles, and its runtime cost for large-scale simulations remains a practical concern.

Reference [8] proposes L2M-AID, a hierarchical multi-agent framework that fuses LLM-based semantic reasoning with Multi-Agent Reinforcement Learning (MARL) for autonomous cyber-physical defense in Industrial IoT environments. Using a Centralized Training, Decentralized Execution paradigm with the MAPPO algorithm, the system achieves a 97.2% detection rate while reducing false positives by over 80% and improving response times fourfold on the SWaT benchmark dataset. The primary limitation lies in the simulation-to-reality gap, as the framework has not yet been validated in live industrial deployments, and adversarial threats such as prompt injection targeting the LLM layer remain unaddressed.

Reference [9] presents a systematic literature review on the application of generative AI and LLMs in cybersecurity, analyzing 42 LLM models across domains including intrusion detection, malware detection, hardware security, and cyber threat intelligence. The review identifies key vulnerabilities in LLMs themselves, including prompt injection, insecure output handling, and data poisoning, while also surveying mitigation strategies such as RLHF, RAG, and QLoRA. The authors note persistent gaps in standardized evaluation benchmarks and highlight computational overhead as a barrier to widespread deployment, calling for more robust and explainable LLM integration in cybersecurity frameworks.

Reference [10] provides a systematic literature review on LLMs in cybersecurity, covering over 300 works encompassing 25 LLMs and more than 10 downstream security scenarios including vulnerability detection, malware analysis, threat intelligence, and penetration testing. The study identifies critical challenges in constructing cybersecurity-oriented LLMs, particularly around data quality, adversarial robustness, and the balance between offensive and defensive applications. Future research directions emphasize the need for domain-specific fine-tuning, improved benchmark standardization, and multimodal LLM integration for comprehensive security coverage.

Reference [11] presents a deep learning-based multi-class intrusion detection system for DDoS attacks in IoT environments, employing CNN, DNN, and Transformer architectures on the CICIoT2023 dataset across binary, three-class, and twelve-class classification tasks. The DNN achieves 99.2% ac-

curacy in binary classification, while SMOTE-based oversampling and log normalization significantly improve performance across imbalanced attack categories. Despite strong results, the system's computational demands and limited evaluation on a single dataset restrict its applicability to heterogeneous, resource-constrained IoT deployments.

Reference [12] introduces MalGEN, a multi-agent generative framework for modeling malicious software by simulating coordinated adversarial workflows, including payload planning, capability selection, and evasion strategy design, within a controlled ethical research environment. Ten novel malware samples synthesized by MalGEN exhibited stealthy characteristics that bypassed leading antivirus and behavioral detection engines, demonstrating the framework's utility in proactive defense preparation and dataset augmentation for training resilient detectors. The primary concern remains the dual-use nature of such a framework, as the underlying techniques could be misused outside controlled research settings despite the ethical safeguards implemented. The reviewed studies collectively illustrate the rapid evolution of AI-driven cybersecurity, spanning multi-agent coordination, LLM-based threat detection, autonomous cyber-physical defense, and generative adversarial simulation. While these approaches demonstrate marked improvements in detection accuracy, contextual reasoning, and response automation, common challenges persist across the literature, including high computational costs, limited real-world validation, adversarial vulnerabilities within the AI systems themselves, and the absence of standardized evaluation benchmarks. Future research must prioritize scalable, explainable, and adversarially robust frameworks that can be deployed reliably in dynamic environments such as smart cities and industrial IoT ecosystems.

III. NOVELTY OF THE WORK

The proposed work introduces a novel AI-driven cybersecurity framework tailored for smart city lighting infrastructure by integrating multi-agent systems with large language model (LLM)-based threat intelligence. Unlike traditional centralized and signature-based security approaches, the proposed system adopts a zone-based architecture, where the city is divided into critical, residential, and commercial zones, enabling context-aware and scalable threat monitoring.

A key novelty lies in the design of a parallel multi-agent framework, where specialized agents—namely the DDoS Detection Agent and Malware Detection Agent—operate concurrently to analyze heterogeneous data streams. This distributed detection mechanism improves response time and enhances system robustness compared to monolithic systems.

Furthermore, the integration of LLM-based threat classification using Groq-powered models introduces adaptive and context-aware decision-making capabilities, allowing the system to identify complex and previously unseen attack patterns. The use of real-time data streaming through Apache Kafka, combined with LangGraph-based agent orchestration, enables continuous data ingestion, processing, and automated response generation in a scalable pipeline.

Another novel aspect of this work is the incorporation of an end-to-end automated response layer, which not only detects threats but also generates mitigation actions such as rate limiting, IP blocking, and device quarantine without human intervention.

Overall, the proposed system distinguishes itself by combining zone-based modeling, multi-agent parallel detection, LLM-driven intelligence, and real-time streaming architecture into a unified framework for securing smart city infrastructure.

IV. DATA GATHERING, MAPPING, AND PREPROCESSING

The proposed cybersecurity framework relies on a structured and labeled dataset specifically designed to evaluate multi-agent threat detection in smart city lighting infrastructure. The dataset consists of two primary components: Distributed Denial-of-Service (DDoS) attack data and malware behavior data, enabling comprehensive analysis of both network-level and device-level threats.

A. Data Gathering

The data used in this study is organized into two categories: DDoS datasets and malware datasets. The DDoS dataset includes multiple attack families such as HTTP Flood, SYN Flood, UDP Flood, Slowloris, DNS Amplification, and volumetric attacks, along with normal traffic samples for baseline evaluation. Similarly, the malware dataset consists of various attack types including ransomware, trojan, botnet, spyware, and rootkit, along with benign device behavior data.

Each dataset contains both malicious and non-malicious instances with explicit ground-truth labels, including is_attack, attack_type, and severity, allowing the system to perform binary classification, multi-class attack identification, and severity-based analysis. The dataset comprises over 1000 records with a balanced mix of attack and normal instances, ensuring realistic evaluation scenarios.

The data reflects real-world smart city conditions by incorporating multiple device identifiers, zone mappings, and diverse event types such as network traffic, device behavior, and system logs. This enables the system to simulate real-time cyber threat scenarios across different zones of the city infrastructure.

B. Data Mapping

Due to the heterogeneous nature of the collected data, a mapping process is applied to standardize and align data from different sources into a unified structure. Separate schemas are defined for DDoS and malware datasets to capture relevant features specific to network traffic and device behavior.

The DDoS dataset schema includes attributes such as source and destination IP addresses, protocol type, packet size, response time, connection type, and status codes. In contrast, the malware dataset captures device-level features such as CPU usage, memory consumption, process execution details, file system changes, and integrity checks.

In addition to structural alignment, each data instance is mapped to a specific zone within the smart city, such as

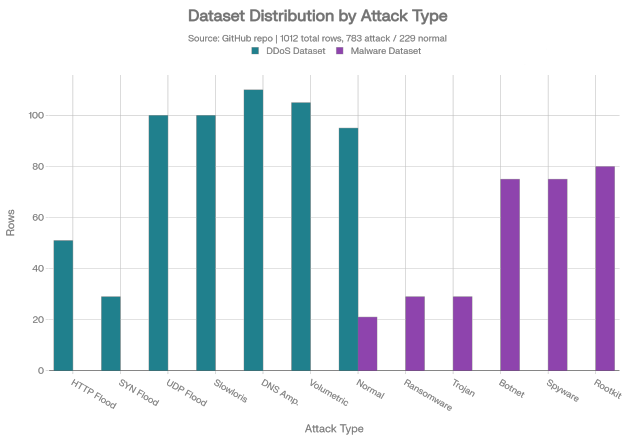

Fig. 1. Distribution of dataset by Attack Type.

critical, residential, or commercial zones. This zone-based mapping enables context-aware threat analysis and allows the detection agents to prioritize threats based on their impact on different regions of the city.

C. Data Preprocessing

The preprocessing stage prepares the mapped data for efficient analysis by the AI agents. This includes data cleaning, validation, normalization, and feature extraction to ensure high-quality input for threat detection.

Data cleaning removes incomplete, inconsistent, or noisy entries, while validation ensures that all records conform to predefined schemas. Normalization techniques are applied to standardize numerical features such as packet size, response time, CPU usage, and memory consumption.

Feature extraction is performed to derive meaningful attributes that represent attack behavior. For DDoS detection, features such as traffic rate, connection frequency, protocol distribution, and latency patterns are utilized. For malware detection, features include abnormal resource usage, suspicious process activity, unauthorized file operations, and integrity check failures.

Additionally, the dataset supports streaming-based evaluation by maintaining temporal ordering through timestamps, enabling realistic simulation of real-time attack detection. This ensures that the preprocessing pipeline aligns with the real-time nature of the proposed multi-agent system.

V. METHODOLOGY

*1) **System Overview**:* The proposed system is an AI-powered multi-agent cybersecurity framework designed to secure smart city lighting infrastructure through real-time threat detection and automated response. The system follows a structured pipeline approach that processes continuous data streams from IoT devices and network sources, enabling efficient monitoring and analysis of potential cyber threats.

At a high level, the framework is organized into a five-layer architecture comprising data sources, data streaming, preprocessing, AI agent-based detection, and response layers. Data generated from IoT-enabled lighting systems, network events, and external threat intelligence feeds is continuously ingested and streamed using a real-time messaging system. This data is then processed and forwarded to specialized detection agents.

The core of the system lies in its multi-agent framework, where independent agents operate in parallel to detect different types of cyber threats. Specifically, a DDoS Detection Agent analyzes network traffic patterns, while a Malware Detection Agent monitors device behavior and anomalies. These agents leverage large language model (LLM)-based intelligence for contextual threat classification and decision-making.

Once a threat is identified, the system triggers automated mitigation actions and generates alerts for monitoring and visualization. This pipeline ensures low-latency detection, scalability, and adaptability, making the proposed solution suitable for dynamic and large-scale smart city environments.

*2) **System Architecture**:* The proposed system follows a layered architecture designed to enable scalable, real-time cybersecurity monitoring for smart city lighting infrastructure. The architecture is composed of five interconnected layers, each responsible for a specific stage in the data processing and threat detection pipeline.

The first layer, Data Sources, consists of IoT-enabled lighting devices, network traffic logs, device-generated events, and external threat intelligence feeds. These sources continuously generate heterogeneous data that reflects the operational and security state of the system.

The second layer, Data Streaming Layer, is responsible for real-time data ingestion and communication. Apache Kafka is utilized as a distributed messaging system to handle high-throughput data streams. Different types of data, such as network events, sensor data, device logs, and cyber alerts, are organized into dedicated topics, ensuring efficient and scalable data flow across the system.

The third layer, Preprocessing Layer, performs initial data refinement before analysis. This includes validation, filtering, and structuring of incoming data to ensure consistency and relevance. The processed data is then forwarded to the intelligent detection layer.

The fourth layer, AI Agent Layer, forms the core of the system. It consists of multiple specialized agents, including the DDoS Detection Agent and the Malware Detection Agent, which operate in parallel. These agents analyze incoming data streams, identify anomalies, and classify potential threats using LLM-based reasoning and predefined detection logic. The use of a multi-agent architecture enhances modularity, scalability, and fault tolerance.

The final layer, Response Layer, is responsible for generating alerts and executing mitigation strategies. Based on the severity and type of detected threat, the system can initiate actions such as rate limiting, IP blocking, or device isolation. Additionally, alerts are forwarded to monitoring dashboards for visualization and further analysis.

This layered architecture ensures efficient data handling, parallel threat detection, and rapid response, making the sys-

tem robust and suitable for deployment in large-scale smart city environments.

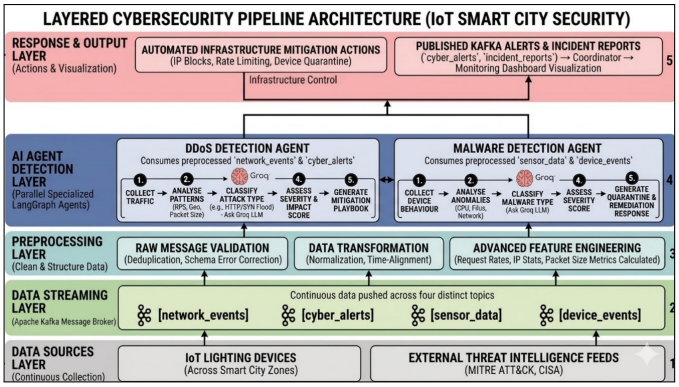

Fig. 2. System Architecture

*3) **Data Streaming Layer**:* The Data Streaming Layer plays a critical role in enabling real-time communication and continuous data flow within the proposed cybersecurity framework. Given the high volume and velocity of data generated by IoT devices and network infrastructure in a smart city environment, an efficient and scalable streaming mechanism is essential. To achieve this, the system leverages Apache Kafka as a distributed event streaming platform.

Apache Kafka is used to ingest, buffer, and distribute data streams across different components of the system in a fault-tolerant and scalable manner. Data from various sources, including network traffic logs, sensor data from smart lighting devices, and system-generated events, is published to Kafka topics. These topics are logically categorized based on the type of data, such as network events, device events, sensor readings, and cybersecurity alerts.

Producers within the system continuously push data into Kafka topics, while consumers—including preprocessing modules and AI agents—subscribe to relevant topics to receive data in real time. This publish-subscribe model ensures loose coupling between system components, allowing independent scaling and improved system resilience.

Additionally, Kafka enables efficient handling of high-throughput data streams and supports real-time analytics by minimizing latency. The streaming layer ensures that incoming data is immediately available for further processing and analysis by the detection agents, thereby facilitating rapid threat identification and response.

Overall, the Data Streaming Layer provides the backbone for real-time data exchange, ensuring reliability, scalability, and low-latency communication across the entire cybersecurity framework.

*4) **AI Agent Framework**:* The AI Agent Framework forms the core intelligence layer of the proposed system, enabling distributed and autonomous threat detection across the smart city infrastructure. Instead of relying on a centralized detection mechanism, the system adopts a multi-agent architecture where specialized agents operate concurrently to analyze different aspects of the incoming data streams.

Each agent is designed to handle a specific category of cyber threat, allowing for modularity and scalability in the system. The agents continuously consume processed data from the streaming layer and perform independent analysis based on their designated roles. This parallel execution significantly reduces detection latency and improves the system's ability to handle large-scale, heterogeneous data in real time.

To coordinate the interaction between multiple agents, the framework utilizes LangGraph for orchestration. LangGraph enables structured communication, task routing, and decision flow between agents, ensuring that relevant information is shared efficiently without creating bottlenecks. It also supports chaining of tasks, allowing agents to collaboratively refine threat analysis when required.

Furthermore, the agents leverage large language model (LLM)-based reasoning to enhance their analytical capabilities. This allows the system to go beyond traditional rule-based detection and identify complex, context-aware attack patterns. The integration of LLMs enables adaptive learning and improves the system's ability to detect previously unseen or evolving threats.

Overall, the AI Agent Framework provides a flexible and intelligent foundation for real-time cybersecurity, combining parallel processing, modular design, and advanced AI-driven decision-making to secure smart city infrastructure effectively.

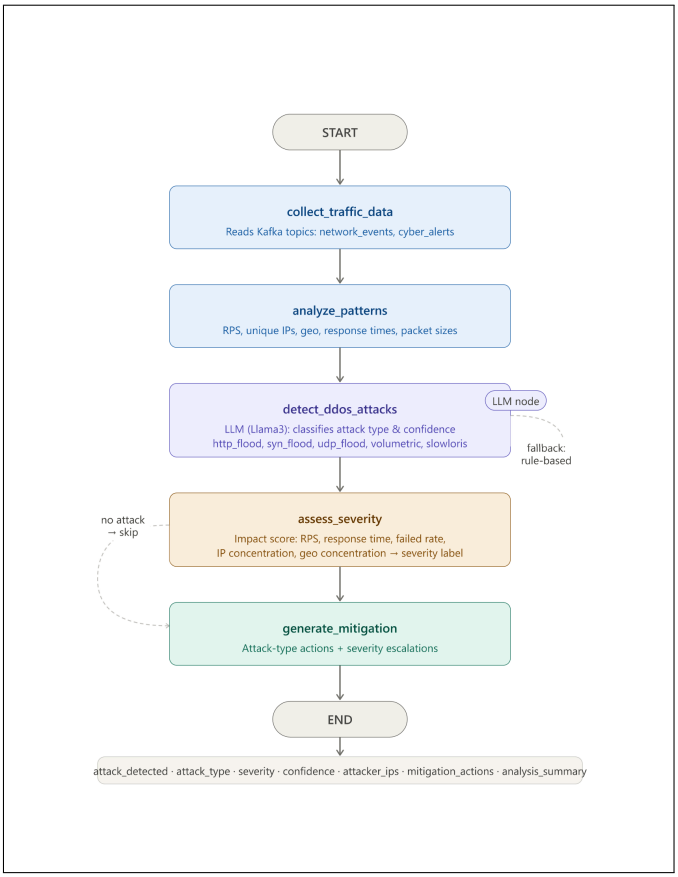

Fig. 3. DDoS Agent Architecture

*5) **DDoS Detection Agent**:* The DDoS Detection Agent is designed to identify and mitigate Distributed Denial-of-Service (DDoS) attacks by continuously monitoring network traffic patterns within the smart city infrastructure. This agent focuses on analyzing high-volume network data streams to detect abnormal spikes, traffic anomalies, and suspicious communication behavior that may indicate an ongoing attack.

The agent consumes real-time network event data from the streaming layer and performs traffic analysis using a combination of statistical methods and AI-driven techniques. Key indicators such as sudden increases in request rates, unusual packet distributions, repeated connection attempts, and abnormal source-destination patterns are evaluated to identify potential DDoS activity.

To enhance detection accuracy, the agent leverages large language model (LLM)-based reasoning for contextual analysis of network behavior. This enables the system to differentiate between legitimate traffic surges and malicious attack patterns, reducing false positives. The agent classifies detected events based on severity levels and assigns risk scores to prioritize response actions.

Once a potential DDoS attack is identified, the agent generates alerts and forwards them to the response layer. It also contributes to automated mitigation by recommending or triggering actions such as rate limiting, traffic filtering, and temporary IP blocking. These responses help maintain system availability and prevent service disruption in critical smart city operations.

Overall, the DDoS Detection Agent provides a robust mechanism for real-time network monitoring and rapid threat identification, ensuring the resilience and reliability of the smart city lighting infrastructure.

*6) **Malware Detection Agent**:* The Malware Detection Agent is responsible for identifying malicious activities and compromised devices within the smart city lighting infrastructure. Unlike network-focused attacks such as DDoS, malware threats often manifest through abnormal device behavior, unauthorized access patterns, and deviations from normal operational profiles. This agent continuously monitors device-level data to detect such anomalies in real time.

The agent consumes device events, sensor data, and system logs from the streaming layer and analyzes behavioral patterns of IoT devices. Key indicators include unusual communication frequency, unauthorized command execution, unexpected configuration changes, and deviations in power or operational metrics. By establishing a baseline of normal device behavior, the agent is able to detect anomalies that may indicate malware infection.

To improve detection capabilities, the Malware Detection Agent incorporates LLM-based contextual analysis, allowing it to interpret complex patterns and correlate multiple indicators of compromise. This enhances its ability to detect sophisticated and previously unseen malware variants that may evade traditional signature-based detection systems.

Upon detecting suspicious activity, the agent classifies the threat, assigns a risk level, and generates alerts for further action. It also contributes to automated mitigation by recommending responses such as device isolation, quarantine, or restricted communication. These actions help prevent the spread of malware across the network and ensure the integrity of the smart city infrastructure.

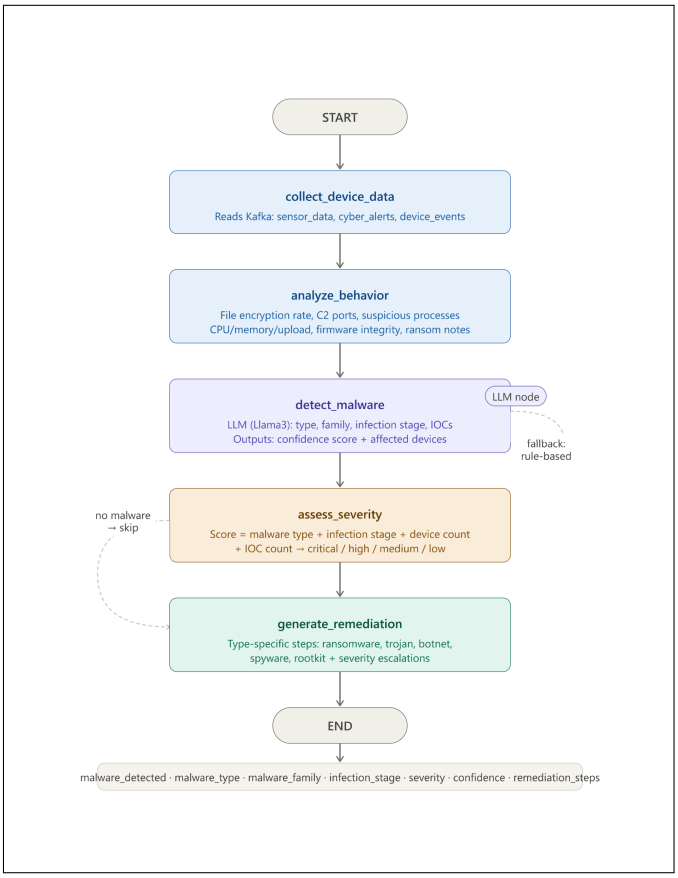

Fig. 4. Malware Agent Architecture

Overall, the Malware Detection Agent enhances the system's ability to detect and respond to device-level threats, providing an additional layer of security alongside network-based detection mechanisms.

*7) **LLM-Based Threat Classification**:* The LLM-Based Threat Classification module enhances the intelligence of the proposed system by enabling context-aware analysis and accurate categorization of detected threats. Traditional rule-based systems often struggle to interpret complex and evolving attack patterns; therefore, the integration of large language models (LLMs) provides a more adaptive and flexible approach to cybersecurity analysis.

In the proposed framework, Groq-powered LLMs are utilized to analyze structured and semi-structured data generated by the detection agents. Once the DDoS Detection Agent or Malware Detection Agent identifies a potential anomaly, relevant features and contextual information are forwarded to the LLM for deeper analysis. The model processes this information to classify the type of attack, assess its severity, and provide reasoning-based insights.

The use of LLMs allows the system to perform semantic understanding of threat patterns rather than relying solely on predefined signatures or thresholds. This capability is particularly useful in identifying zero-day attacks and multi-vector threats, where traditional methods may fail. Additionally, LLMs assist in correlating events across different data sources, improving the overall accuracy and reliability of threat detection.

Another advantage of this approach is the ability to generate human-readable explanations for detected threats, which can assist system administrators in understanding the nature of the attack and appropriate actions. This interpretability enhances trust and usability in real-world deployments.

Overall, the integration of LLM-based threat classification significantly improves the system's ability to detect, interpret, and respond to complex cybersecurity threats in a dynamic smart city environment.

*8) **Response and Mitigation Layer**:* The Response and Mitigation Layer is responsible for taking appropriate actions once a potential threat has been detected and classified by the AI agents. This layer ensures that the system not only identifies cyber threats but also responds to them in a timely and effective manner, minimizing potential damage to the smart city infrastructure.

Based on the severity and type of threat identified by the detection agents and the LLM-based classification module, the system triggers automated mitigation strategies. For network-based attacks such as DDoS, actions may include rate limiting, traffic filtering, and temporary blocking of suspicious IP addresses. In the case of malware-related threats, the system can initiate device isolation, restrict unauthorized communication, or quarantine compromised nodes to prevent further propagation.

The response mechanism is designed to operate in real time, ensuring minimal delay between detection and action. Alerts and mitigation decisions are published to dedicated messaging topics within the streaming layer, enabling seamless communication between system components. This also allows monitoring systems and dashboards to visualize incidents and track system status.

Additionally, the layer supports extensibility by allowing new mitigation strategies to be incorporated as the threat landscape evolves. The combination of automated response and real-time alerting enhances the resilience of the system, ensuring continuous operation of critical smart city services.

Overall, the Response and Mitigation Layer transforms the system from a passive detection framework into an active defense mechanism, capable of autonomously protecting smart city infrastructure against cyber threats.

*9) **Monitoring and Visualization**:* The Monitoring and Visualization component provides a centralized interface for observing system activity, analyzing detected threats, and evaluating the overall security posture of the smart city infrastructure. This layer enables real-time visibility into the operations of the multi-agent framework and supports informed decision-making for system administrators.

All alerts, threat classifications, and mitigation actions generated by the response layer are aggregated and presented through a monitoring dashboard. The dashboard displays key information such as detected attack types, severity levels, affected zones (critical, residential, or commercial), and response actions taken. This allows administrators to quickly understand the current state of the system and identify potential vulnerabilities.

In addition to real-time monitoring, the system maintains logs of historical events, enabling trend analysis and performance evaluation. Visualization tools can be used to analyze patterns in cyberattacks, such as frequency, origin, and impact across different zones. This information is valuable for improving future detection strategies and enhancing system robustness.

The monitoring layer also acts as a coordination interface, allowing administrators to intervene if necessary, review automated decisions, and adjust system parameters. By providing both real-time insights and historical analysis, this component ensures transparency, accountability, and continuous improvement of the cybersecurity framework.

Overall, the Monitoring and Visualization layer enhances the usability and effectiveness of the system by transforming complex security data into actionable insights, thereby supporting efficient management of smart city cybersecurity operations.

VI. Experimental Results

The proposed multi-agent cybersecurity framework was evaluated using multiple large language models (LLMs) across diverse attack scenarios involving both DDoS and malware datasets. The evaluation was conducted using key performance metrics, including accuracy, precision, recall, F1-score, false positive rate (FPR), average confidence, MITRE ATTCK TTP coverage, and time-to-detect (TTD), providing a comprehensive assessment of detection performance and system reliability. Among the evaluated models, LLaMA-3.1-70B demonstrated the best overall performance, achieving the highest accuracy and F1-score while maintaining zero false positives and strong prediction confidence. Other models such as Mistral-7B and Mixtral-8x7B also showed comparable performance, highlighting the robustness of the proposed system across different LLMs.

*a) **LLM Performance Comparison (Accuracy, Precision, Recall, F1 Score)**:* This section presents a comparative analysis of multiple large language models (LLMs) evaluated on the proposed multi-agent cybersecurity framework using standard classification metrics, including accuracy, precision, recall, and F1-score.

The experimental results demonstrate that high-capacity models such as LLaMA-3.1-70B, Mistral-7B, and Mixtral-8x7B consistently achieve superior performance across all evaluation metrics. These models attain an accuracy of 93.8%, precision of 100.0%, recall of 93.3%, and F1-score of 96.6%, indicating a well-balanced performance with minimal false positives and strong detection capability. The perfect precision

across these models highlights the system's ability to avoid misclassifying benign traffic as malicious, which is critical in real-world smart city deployments where false alarms can lead to unnecessary disruptions.

In contrast, mid-sized models such as Qwen-32B and Gemma-9B exhibit slightly lower performance, with accuracy around 87.5% and F1-scores ranging between 92.9% and 93.3%. While these models still perform reliably, the reduction in recall suggests a decreased ability to detect certain attack patterns, particularly in more complex or multi-vector scenarios.

The performance gap becomes more pronounced with smaller models such as LLaMA-3.1-8B, which achieves only 68.8% accuracy and an F1-score of 80.0%. This indicates that lower-capacity models struggle to generalize across diverse cybersecurity scenarios, likely due to limited contextual understanding and reduced representation capability.

Overall, the results confirm that model size and capability play a significant role in cybersecurity applications. Larger models are better suited for capturing complex attack patterns and maintaining high detection accuracy, making them more appropriate for deployment in critical smart city environments.

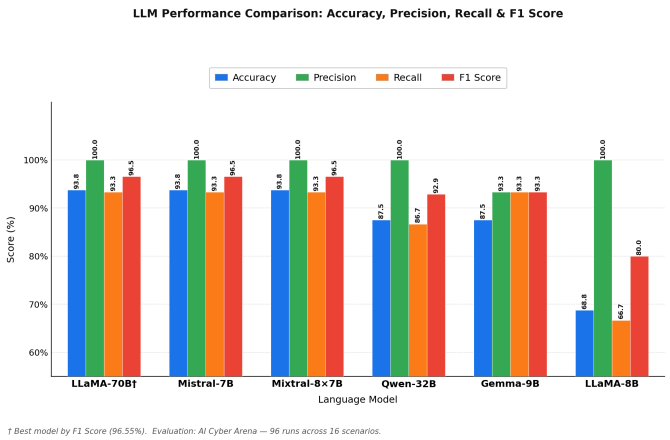

Fig. 5. Comparison of LLM performance across key metrics including accuracy, precision, recall, and F1-score for DDoS and malware detection.

*b) **Evaluation Heatmap Analysis**:* The evaluation heatmap provides a consolidated visualization of model performance across multiple metrics, including accuracy, precision, recall, F1-score, average confidence, and MITRE ATTCK TTP coverage. This representation enables a holistic comparison of models and highlights performance consistency across different dimensions.

From the heatmap, it is evident that most models achieve near-perfect precision (close to 100%), indicating that the system is highly effective in minimizing false positives. This is particularly important in cybersecurity systems where false alerts can overwhelm monitoring systems and reduce operational efficiency.

However, variations are observed in recall and average confidence scores. High-capacity models such as LLaMA-3.1-70B and Mixtral-8x7B maintain consistently high recall values, demonstrating their ability to detect a wide range of attack scenarios. In contrast, smaller models exhibit reduced recall, suggesting missed detections in certain cases.

The heatmap also reveals differences in average confidence levels, with larger models showing higher confidence in their predictions. This indicates stronger certainty and reliability in decision-making. Additionally, TTP coverage values remain relatively moderate across most models, highlighting an area for potential improvement in mapping detected threats to a broader range of adversarial techniques.

Overall, the heatmap effectively illustrates that while most models perform well in classification tasks, differences emerge in their ability to generalize, interpret, and confidently classify complex threats.

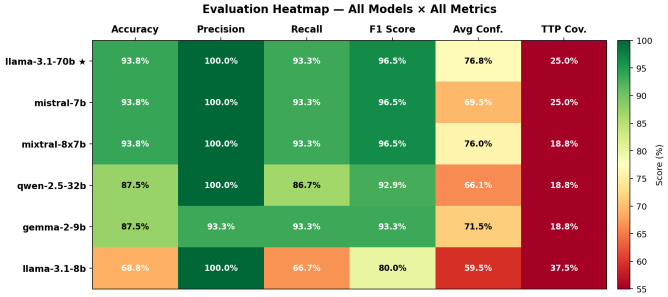

Fig. 6. Heatmap visualization of model performance across multiple evaluation metrics, highlighting variations in accuracy, precision, recall, F1-score, confidence, and TTP coverage.

*c) **F1 Score vs Accuracy Comparison**:* This section analyzes the relationship between F1-score and accuracy to evaluate the balance between precision and recall across different models.

The results show that high-performing models such as LLaMA-3.1-70B, Mistral-7B, and Mixtral-8x7B achieve both high accuracy (93.8%) and high F1-scores (96.6%), indicating a strong balance between minimizing false positives and maximizing detection rates. This balance is crucial in cybersecurity applications, where both types of errors—false positives and false negatives—can have significant consequences.

The comparison also highlights that accuracy alone is not sufficient to evaluate model performance. For instance, models with similar accuracy values may exhibit differences in F1-score, reflecting variations in recall. This is particularly evident in mid-tier models, where slight drops in recall lead to lower F1-scores despite relatively high accuracy.

Lower-capacity models such as LLaMA-3.1-8B show a significant decline in both accuracy and F1-score, indicating poor overall performance and an inability to effectively capture complex attack patterns. The gap between these metrics further emphasizes the limitations of smaller models in real-time cybersecurity applications.

Overall, this analysis reinforces the importance of using F1-score alongside accuracy as a primary evaluation metric, as it provides a more comprehensive measure of model effectiveness in imbalanced and high-risk environments.

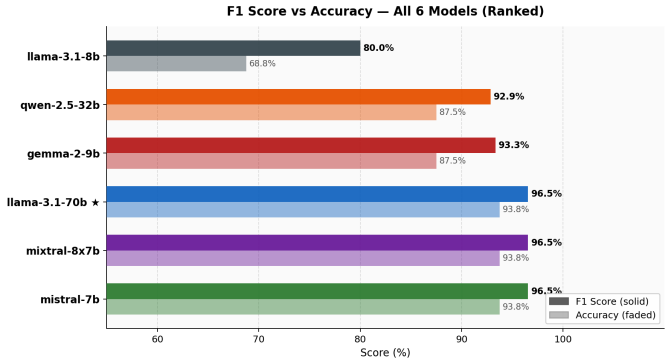

Fig. 7. Comparison of F1-score and accuracy across different LLMs, illustrating the balance between precision and recall in threat detection.

*d) **Average Confidence and TTP Coverage Analysis:*** This section evaluates model performance in terms of prediction confidence and coverage of MITRE ATTCK techniques, providing insights into interpretability and threat mapping capabilities.

The results indicate that models such as LLaMA-3.1-70B and Mixtral-8x7B achieve the highest average confidence scores (approximately 76%), suggesting that these models produce more reliable and consistent predictions. Higher confidence levels are particularly important in automated systems, as they directly influence decision-making and response actions.

In contrast, smaller models exhibit lower confidence scores, indicating uncertainty in predictions and reduced reliability. This can lead to inconsistent detection outcomes and decreased trust in automated mitigation strategies.

The analysis of TTP coverage reveals that most models achieve moderate coverage levels (18–25%), indicating that while they are effective at detecting attacks, their ability to map these attacks to a comprehensive set of adversarial techniques is limited. Interestingly, the LLaMA-3.1-8B model achieves higher TTP coverage (37.5%), suggesting that it identifies a broader range of attack patterns, albeit with lower accuracy and higher false positive rates.

This observation highlights a key trade-off between precision and coverage, where models with broader detection capabilities may sacrifice classification accuracy. Balancing these aspects is essential for developing robust and interpretable cybersecurity systems.

*e) **Model Ranking and Performance Summary:*** This section presents a consolidated ranking of all evaluated models based on multiple performance metrics, including accuracy, precision, recall, F1-score, false positive rate (FPR), average confidence, TTP coverage, and time-to-detect (TTD).

The results clearly indicate that LLaMA-3.1-70B is the best-performing model, achieving the highest overall performance with an accuracy of 93.8%, F1-score of 96.6%, and zero false positive rate. Its high confidence level and balanced performance across all metrics make it the most suitable model for deployment in critical smart city environments.

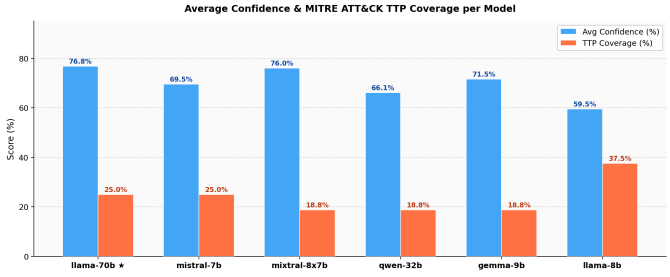

Fig. 8. Comparison of average prediction confidence and MITRE ATTACK TTP coverage across models, showing the trade-off between prediction certainty and threat mapping capability.

Models such as Mistral-7B and Mixtral-8x7B closely follow, demonstrating comparable performance and offering potential alternatives where computational resources are limited. These models maintain high accuracy and F1-scores while slightly compromising on confidence and coverage.

Mid-tier models, including Gemma-9B and Qwen-32B, provide moderate performance and may be suitable for less critical applications. However, their lower recall and confidence levels indicate reduced effectiveness in complex threat scenarios.

The lowest-ranked model, LLaMA-3.1-8B, shows significantly lower accuracy and higher false positive rates, making it less suitable for deployment in real-time cybersecurity systems.

An important observation is that all models exhibit a consistent time-to-detect (TTD) of approximately 28.2 seconds, highlighting the efficiency and stability of the proposed multi-agent architecture in handling real-time data streams.

Overall, the ranking validates the effectiveness of the proposed system and emphasizes the importance of selecting appropriate models based on performance requirements and resource constraints.

Rank	Model	Accuracy	Precision	Recall	F1	FPR	TTP Cov.	Avg Conf.	TTD
1 (BEST)	llama-3.1-70b	93.8%	100.0%	93.3%	96.6%	0.0%	25%	76.8%	28.2s
2	mistral-7b	93.8%	100.0%	93.3%	96.6%	0.0%	25%	69.5%	28.2s
3	mixtral-8x7b	93.8%	100.0%	93.3%	96.6%	0.0%	19%	76.0%	28.2s
4	gemma-2-9b	87.5%	93.3%	93.3%	93.3%	100.0%	19%	71.5%	28.2s
5	qwen-2.5-32b	87.5%	100.0%	86.7%	92.9%	0.0%	19%	66.1%	28.2s
6	llama-3.1-8b	68.8%	100.0%	66.7%	80.0%	38.0%	19%	59.5%	28.2s

Fig. 9. Overall ranking of LLMs based on multiple evaluation metrics including accuracy, F1-score, false positive rate, confidence, and time-to-detect.

VII. CONCLUSION

This paper presented an AI-powered multi-agent cybersecurity framework designed to secure smart city lighting infrastructure against evolving cyber threats. By integrating a zone-based architecture with a distributed multi-agent system, the proposed approach enables real-time detection and

response to both network-level attacks, such as Distributed Denial-of-Service (DDoS), and device-level threats, including malware intrusions. The use of Apache Kafka for real-time data streaming, LangGraph for agent orchestration, and large language models (LLMs) for intelligent threat classification ensures a scalable, adaptive, and efficient security solution.

Experimental results demonstrate that the proposed system achieves high detection accuracy, low false positive rates, and consistent real-time performance across multiple LLMs. Among the evaluated models, LLaMA-3.1-70B achieved the best overall performance, maintaining a strong balance between precision and recall while ensuring reliable threat detection. The results also highlight the effectiveness of combining multi-agent architectures with LLM-based reasoning to handle complex and multi-vector cyberattacks in dynamic environments.

Furthermore, the system demonstrates robustness and flexibility, as it performs consistently across different models and attack scenarios while maintaining stable detection latency. The incorporation of severity-based classification and automated response mechanisms enhances the practical applicability of the framework in real-world smart city deployments.

In conclusion, the proposed approach provides a comprehensive and scalable solution for securing IoT-based smart city infrastructure. Future work may focus on expanding the system to support additional attack types, improving TTP coverage through enhanced threat intelligence integration, and deploying the framework in real-world environments for large-scale validation.

REFERENCES

[1] B. Chen et al., "Multi-Agent AI Systems for Coordinated Cybersecurity in Smart Cities," *International Journal for Research Publication and Seminar*, vol. 16, no. 3, pp. 4–22, Jul.–Sep. 2025. doi: 10.36676/jrps.v16.i3.287.

[2] R. Molleti, V. Goje, P. Luthra, and P. Raghavan, "Automated Threat Detection and Response Using LLM Agents," *World Journal of Advanced Research and Reviews*, vol. 24, no. 2, pp. 79–90, 2024. doi: 10.30574/wjarr.2024.24.2.3329.

[3] K. Kurniawan, E. Kiesling, and A. Ekelhart, "CyKG-RAG: Towards Knowledge-Graph Enhanced Retrieval Augmented Generation for Cybersecurity," in *Proc. International Workshop on Retrieval-Augmented Generation Enabled by Knowledge Graphs (RAGE-KG 2024), co-located with ISWC 2024*, Baltimore, USA, Nov. 2024, *CEUR Workshop Proceedings*, vol. 3950, pp. 51–64.

[4] M. Houichi, F. Jaidi, and A. Bouhoula, "Cyber Security within Smart Cities: A Comprehensive Study and a Novel Intrusion Detection-Based Approach," *Computers, Materials & Continua*, vol. 81, no. 1, pp. 393–441, Oct. 2024. doi: 10.32604/cmc.2024.054007.

[5] Y. Hmimou, M. Tabaa, A. Khiat, and Z. Hidila, "A Multi-Agent System for Cybersecurity Threat Detection and Correlation Using Large Language Models," *IEEE Access*, vol. 13, 2025. doi: 10.1109/ACCESS.2025.3602681.

[6] A. Bhardwaj, S. Godika, and Y. Loonker, "MALCDF: A Distributed Multi-Agent LLM Framework for Real-Time Cyber Defense," arXiv preprint arXiv:2512.14846, Dec. 2025.

[7] J. Yu, X. Xie, Q. Hu, Y. Ma, and Z. Zhao, "Chimera: Harnessing Multi-Agent LLMs for Automatic Insider Threat Simulation," in *Proc. Network and Distributed System Security (NDSS) Symposium 2026*, San Diego, CA, USA, 2026. arXiv:2508.07745.

[8] T. Xu, Z. Wen, X. Zhao, J. Wang, Y. Li, and C. Liu, "L2M-AID: Autonomous Cyber-Physical Defense by Fusing Semantic Reasoning of Large Language Models with Multi-Agent Reinforcement Learning," arXiv preprint arXiv:2510.07363, Oct. 2025.

[9] M. A. Ferrag, F. Alwahedi, A. Battah, B. Cherif, A. Mechri, N. Tihanyi, T. Bisztray, and M. Debbah, "Generative AI in Cybersecurity: A Comprehensive Review of LLM Applications and Vulnerabilities," *Internet of Things and Cyber-Physical Systems*, Feb. 2025. doi: 10.1016/j.iotcps.2025.01.004.

[10] J. Zhang, H. Bu, H. Wen, Y. Liu, H. Fei, R. Xi, L. Li, Y. Yang, H. Zhu, and D. Meng, "When LLMs Meet Cybersecurity: A Systematic Literature Review," *Cybersecurity*, vol. 8, no. 1, article 55, Feb. 2025. doi: 10.1186/s42400-025-00361-w.

[11] S. A. Wahab, S. Sultana, M. Mujahid, J. Khan, J. Ali, and A. Mylonas, "A Multi-Class Intrusion Detection System for DDoS Attacks in IoT Networks Using Deep Learning and Transformers," *Sensors*, vol. 25, no. 15, article 4845, Aug. 2025. doi: 10.3390/s25154845.

[12] B. Saha and S. K. Shukla, "MalGEN: A Generative Agent Framework for Modeling Malicious Software in Cybersecurity," arXiv preprint arXiv:2506.07586, Jun. 2025.